Logic Gates / Authentication – Exhibition Quiz

Project: AuthFlow & Logic Lab | **Exhibition:** 14 November

How to use this quiz: cover the answers, say your answer aloud, then check. Redo any question you miss. Score yourself at the end of each section.

Section 1: Logic Gates (10 questions)

1. Which gate gives output 1 **only when both inputs are 1**?
2. Which gate gives output 1 when **at least one** input is 1?
3. What does a NOT gate do to its input?
4. What is the output of a NAND gate when A = 1 and B = 1?
5. What is the output of a NOR gate when A = 0 and B = 0?
6. What is the output of an XOR gate when A = 1 and B = 1?
7. What is the output of an XNOR gate when A = 0 and B = 0?
8. What is the output of an OR gate when A = 0 and B = 1?
9. Which two gates are the "opposites" of AND and OR?
10. Which gate gives 1 when the two inputs are **different**?

Answers (Section 1)

1. AND
2. OR
3. It flips (inverts) the input: 0 becomes 1, 1 becomes 0.
4. 0 (NAND is AND flipped)
5. 1 (NOR is OR flipped)
6. 0 (the inputs are the same)
7. 1 (the inputs are the same)
8. 1
9. NAND (opposite of AND) and NOR (opposite of OR)
10. XOR

Section 2: Truth Table Drill

Fill in the output for each row, then check below.

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	?	?	?	?	?	?
0	1	?	?	?	?	?	?
1	0	?	?	?	?	?	?
1	1	?	?	?	?	?	?

Answer table

A	B	AND	OR	NAND	NOR	XOR	XNOR
0	0	0	0	1	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	1	0	0	0	1

Memory trick: NAND, NOR and XNOR are always the exact opposite of AND, OR and XOR in the same row.

Section 3: Number Systems (8 questions)

1. What is the base of binary, octal, decimal and hexadecimal?
2. Which digits and letters does hexadecimal use?
3. Convert binary **1010** to decimal.
4. Convert decimal **45** to binary.
5. Convert decimal **8** to octal.
6. Convert decimal **255** to hexadecimal.
7. Convert hexadecimal **1F** to decimal.
8. Why do computers use binary?

Answers (Section 3)

1. Binary 2, octal 8, decimal 10, hexadecimal 16.

2. Digits 0 to 9 and letters A to F (A = 10, B = 11, C = 12, D = 13, E = 14, F = 15).
3. 10 (8 + 2)
4. 101101
5. 10 (octal)
6. FF
7. 31 (1 x 16 + 15)
8. Electronic circuits have two clear states, on and off, which map naturally to 1 and 0. This is more reliable than many voltage levels.

Quick reference for 45: Binary 101101, Octal 55, Decimal 45, Hexadecimal 2D.

Section 4: My Physical Circuits (8 questions)

1. In my AND circuit, how are switches A and B connected?
2. In my OR circuit, how are switches A and B connected?
3. In my NOT circuit, where is the switch connected, and what happens when it is closed?
4. In the circuit, what does the LED represent?
5. What do the switches represent?
6. In the AND circuit, when does the LED glow?
7. In the OR circuit, when does the LED glow?
8. Why is a resistor connected in series with each LED?

Answers (Section 4)

1. **In series** (one after the other, in one path).
2. **In parallel** (two separate paths side by side).
3. The switch is connected **across the LED branch**. When it is closed, the current bypasses the LED, so the LED turns off (input 1 gives output 0). When it is open, the current flows through the LED, so it is on (input 0 gives output 1).
4. The LED is the **output Y**: lit means 1, off means 0.
5. The switches are the **inputs**: closed means 1, open means 0.
6. Only when **both switches are closed**, because the current needs a complete path.
7. When **either or both** switches are closed, because either path completes the circuit.
8. It **limits the current** so the LED is not damaged by the battery voltage.

Section 5: AuthFlow – Authentication (10 questions)

1. What is authentication?
2. Which service handles authentication in AuthFlow?
3. What does the email validation check?
4. Why is there a show/hide password option?
5. What is a session?
6. What happens after a successful login?
7. How does the forgot password feature work?
8. What does Google sign-in let a user do?
9. Which languages did you use to build the website?
10. Where is the website hosted?

Answers (Section 5)

1. Verifying that a user is who they claim to be, usually with an email and a password.
2. **Supabase**, which manages authentication, sessions and password recovery.
3. That the email has a proper format: an @ symbol and a dot in the domain.
4. So users can check what they typed and avoid mistakes. It improves usability.
5. A way for the website to remember that a user is logged in without asking for the password on every page.
6. The user is redirected to the dashboard, which welcomes them and shows basic account information.
7. The user asks for a reset link. A link is sent to their email, and they set a new password through it.
8. Sign in with an existing Google account instead of creating a separate password.
9. HTML, CSS and JavaScript.
10. On **Vercel**, with the code maintained on GitHub.

Section 6: Judge-Style Viva Questions

Practise answering each in 2 to 3 sentences.

1. Where are logic gates used in real life?
2. What is the difference between OR and XOR?
3. Why did you choose to combine web development and digital electronics?

4. Are passwords stored safely?
5. What was the hardest part of the project?
6. What would you add if you had more time?
7. Why do your LED circuits need a resistor?
8. How can the same gate be built with electronics and with software?

Model answers (use your own words)

1. Inside CPUs, memory chips, calculators, alarms and every digital device.
2. OR gives 1 when at least one input is 1, including 1 and 1. XOR gives 0 when both are 1, because it needs the inputs to be different.
3. Both are about how computers work: one shows how websites keep users secure, the other shows how computers think and calculate.
4. Authentication is handled by Supabase, which stores passwords in a hashed form, not as plain text.
5. Answer honestly in your own words, for example getting Google sign-in or the password reset flow working.
6. For example more gates and circuits, binary arithmetic, or user profile editing.
7. It limits the current through the LED. Without it, the battery can damage the LED.
8. The simulator in my website calculates the gate output in code, while my circuits show the same logic using real switches and LEDs.

Score Sheet

Section	Score
1. Logic gates	___ / 10
2. Truth tables	___ / 4 rows
3. Number systems	___ / 8
4. Physical circuits	___ / 8
5. AuthFlow	___ / 10

Target: 90% or higher by the evening, then one final run on the morning of the 14th.